UUM
Universiti Utara Malaysia

FINAL EXAMINATION
SECOND SEMESTER 2023/2024 SESSION

COURSE CODE/NAME	:	STIK2044 / OPERATING SYSTEM
DATE	:	25 JULY 2024 (THURSDAY)
TIME	:	9.00 A.M. – 11.30 A.M. (2 ½ HOURS)
VENUE	:	DP1, SOIS, INASIS TM

INSTRUCTIONS :

1. This examination paper has **THREE (3) SECTIONS: A, B,** and **C**, in **TEN (10)** printed pages excluding the cover page. **SECTION A** has **TWENTY (20)** questions, **SECTION B** has **NINE (9)** questions and **SECTION C** has **(5)** questions.
2. Answer **ALL** questions in **SECTION A, B** and **C** in the space provided.
3. You are **NOT ALLOWED** to remove the examination paper from the examination hall.

MATRIC NO.: _____ (in words) | | | | | | | (in numbers)

IDENTIFICATION CARD NO.: | | | | | | | | | | | | |

LECTURER NAME: _____

GROUP: | | **TABLE NO.:** | | | | |

DO NOT OPEN THIS EXAMINATION PAPER UNTIL INSTRUCTED

SECTION A: SHORT ANSWER QUESTIONS (20 MARKS)

INSTRUCTION: Fill in the blanks with the correct answer based on the options provided in the box below:

parallel	preemptive	compression	path	traffic
integrity	device management	dynamic partition	relocatable dynamic partitions	memory manager
address	dining philosophers	user interface	starvation	readers-writers
volume	waiting	mutual exclusion	working set	turnaround

1. The _____ is the part of the operating system that users directly interact with, and it is one of its most unique components.

2. Effective _____necessitates that this operating system segment uniquely identifies each device, initiates its operation when suitable, monitors its progress, and ultimately deallocates the device to make the operating system available for the next waiting process.

3. The _____ verifies each memory space request and allocates an unused portion of memory if the request is valid.

4. In the _____ allocation scheme, memory is assigned to an incoming job in a single contiguous block, with each job precisely receiving the amount of memory it requests upon being loaded for processing.

5. With the _____ allocation scheme, the Memory Manager relocates programs to consolidate all the empty blocks and compress them to form a single block of memory large enough to accommodate some or all of the jobs queued for entry.

6. One innovation that enhanced the performance of demand paging schemes was the introduction of the _____, defined as the set of pages present in memory that can be directly accessed without resulting in a page fault.

7. Once a system is breached, the _____ of every file on the system, along with the data contained in those files, can no longer be trusted.

8. Priority scheduling can lead to the problem of _____ if aging is not implemented.

9. The total time taken from the submission of a process to its completion is called _____ time.

10. A _____ scheduling algorithm can preempt a running process if a higher priority process arrives.
11. The _____ time is the total time a process spends in the ready queue before getting executed.

12. The _____ problem is used to illustrate synchronization issues where a limited number of resources are shared among processes.

13. The _____ problem is a classic synchronization problem used to illustrate the challenges of allocating resources among several processes.

14. The primary goal of process synchronization is to ensure that multiple processes can operate in _____ without interfering with each other.

15. _____ is a technique used to ensure that no two processes are in their critical section at the same time.

16. A procedure used to reduce the amount of space required to store data by reducing, encoding, or abbreviating repetitive terms or characters is known as data _____.

17. The modules of the I/O subsystem that monitors the status of every device, control unit, and channel are known as I/O _____ controller.

18. Master file directory (MFD) is a file stored immediately after the _____ descriptor that lists the names and characteristics of every file contained in that volume.

19. In a direct file organization environment, relative _____ indicates the position of a record relative to the beginning of the file.

20. _____ is the sequence of directories and subdirectories the operating system must follow to find a specific file.

SECTION B: STRUCTURED QUESTIONS (30 MARKS)
INSTRUCTION: Answer **ALL** the questions in the space provided.

1. Regardless of the setup of the Operating System (OS), list down **FOUR (4)** minimum tasks that keep the OS working smoothly.

(4 marks)

2. Memory partitions may be allocated based on first-fit memory or best-fit memory allocation. State **ONE (1)** difference between first-fit and best-fit memory allocation.

(2 marks)

3. One of the ways to overcome the threat is by applying system protection. Elaborate **FOUR (4)** systems protection methods that can be applied.

(8 marks)

4. The processes may be pre-empted when they share a single Central Processing Unit (CPU). What happens to a process when it is pre-empted? **(2 marks)**

5. Define the critical section. **(2 marks)**

6. Discuss the purpose of process synchronization in operating systems. **(2 marks)**

7. The Universal Serial Bus (USB) controller assigns bandwidth to each device depending on its priority. Write the level of priority (either HIGH, MEDIUM, or LOW) for the following situations.

(3 marks)

SITUATIONS	PRIORITY LEVEL
Bulk transfers or exchanges that can accommodate slower data flow such as a file update.	
Real-time exchanges where no interruption in the data flow is allowed, such as video or sound data.	
Devices that can allow occasional interrupts without jeopardizing the use of the device, such as a keyboard or joystick.	

8. State **ONE (1)** similarity and **ONE (1)** main difference between inter-record gap (IRG) and inter-block gap (IBG). **(3 marks)**

Similarity:	

Difference:	**IRG**	
	IBG	

9. State the seek strategy that is appropriate to use in the following situations.

(4 marks)

Situation	Seek Strategy
It works well with moderate loads, but it has the problem of localization under heavy loads.	
It works well with light to moderate loads, and it eliminates the problem of indefinite postponement.	
It works well with light loads, but as soon as the load grows, service time becomes unacceptably long.	
It works well with moderate to heavy loads, and it has a very small variance in service times.	

SECTION C: PROBLEM / COMPUTATIONAL QUESTIONS (50 MARKS)
INSTRUCTION: Answer **ALL** the questions in the space provided.

1. There are four essential managers of every major operating system: The Memory Manager, Processor Manager, Device Manager, and File Manager. Draw a diagram to illustrate how each of these managers are connected with each other and connected with the User Interface.

(4 marks)

2. Given a fixed-partition memory table and Job List table.

Partition #	Partition Size	Memory Size	Job	Size of Job
1	100k	200k	1	30K
2	25k	300k	2	50K
3	25k	325k	3	30K
4	50k	350k	4	20K

Based on the given table,

a) Which Job is in the Partition 1? Any internal fragmentation occurred? If yes, how big the internal fragmentation is?

(3 marks)

b) Which job is in Partition 2? Why?

(2 marks)

c) Which job is in Partition 3?

(1 mark)

3. Given the following table of processes with their respective arrival times and burst times.

Process ID	Arrival Time	Burst Time
P1	0	8
P2	1	6
P3	2	9
P4	3	5
P5	4	2
P6	5	4
P7	6	2
P8	7	6

a) Use the Shortest Remaining Time First (SRTF) scheduling algorithm to present the sequence of execution and processing times in a Gantt chart.

(12 marks)



b) Fill in the table with completion time, waiting time and turnaround time for the processes.

(8 marks)

Process ID	Arrival Time	Burst Time	Completion Time	Turnaround Time (TAT)	Waiting Time (WT)
P1	0	8			
P2	1	6			
P3	2	9			
P4	3	5			
P5	4	2			
P6	5	4			
P7	6	2			
P8	7	6			

4. Given that it takes 1.75ms to travel from one track to the next of a hard drive; that the arm is originally positioned at Track 15 moving toward the low-numbered tracks; and that you are using the LOOK scheduling policy. Compute the total seek time to satisfy the following requests — 04, 40, 35, 11, 14, and 07. Assume all requests are initially present in the wait queue. (Ignore rotational time and transfer time; just consider seek time.)

(10 marks)

Head Path	No. of Tracks Travelled	Seek Time (ms)
15 → 14	1	1 x 1.75 = 1.75
TOTAL SEEK TIME		

5. The table below shows a list of access controls on 5 files.

File	Access
File 1	USER1 (1111), USER2 (1010), USER4 (1110), USER5 (0010), WORLD (0000)
File 2	USER2 (1010), USER3 (1010), USER4 (0010), WORLD (0000)
File 3	USER2 (1111), USER4 (0010), WORLD (0000)
File 4	USER1 (1010), USER5 (1111), WORLD (0000)
File 5	USER5 (1111), WORLD (0000)

If Read Access, Write Access, Execute Access, and Delete Access are represented using binary code in order of 4 bits, solve the following problem:

a) List the file names and users who can only access the file for READ and EXECUTE.

(4 marks)

b) If the owner of the file has all access rights to the file, who is the owner of each file listed above.

(5 marks)

c) What does WORLD (0000) mean in the access control above?

(1 mark)

END OF QUESTIONS